

From Competing Pension Regimes to a Pillar System:

Informality, Transitional Dynamics, and Distributional Welfare in
Colombia

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Abstract

This paper studies Colombia's counterfactual transition from the competing pension regimes created by Law 100 of 1993 to the pillar architecture of Law 2381 of 2024. I embed the reform in a two-period overlapping-generations economy with heterogeneous workers, endogenous informality, PAYG eligibility, portable funded balances, fiscal closure, and general-equilibrium prices. Under Law 100, workers choose among informality, capitalization, and PAYG. Under the pillar system, formal workers no longer choose an exclusive regime: earnings up to 2.3 minimum wages enter PAYG and only the excess enters an individual account. The legal mapping assigns 13.2 percent of that excess to the funded account, applies the statutory declining replacement formula and one-minimum-wage floor, and calibrates the solidarity benefit to the 2023 extreme-poverty line. The Law 100 benchmark produces 50.9 percent informality, 46.0 percent funded participation, and 3.1 percent PAYG. The central pillar steady state has about 77.5 percent informality, 6.2 percent PAYG-only formality, and 16.3 percent mixed formality, with a 53.4 percent consumption tax. The transition adds public debt and a bounded fiscal rule that responds contemporaneously to the tax required to stabilize debt while retaining a small long-run tax anchor. Twenty-five percent of public debt crowds out domestic capital. Under this closure, the large two-generation oscillation generated by period-by-period budget balance disappears: tax, informality,

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capital, and debt converge to the pillar steady state. The first reform cohort loses 5.5 percent in consumption-equivalent welfare, and the stationary loss is 3.3 percent, with substantial dispersion within cohorts. The exercise is a model-based counterfactual, not an evaluation of an implemented reform: the integral entry into force of Law 2381 remains suspended.

Keywords: pension reform; informality; pillar system; PAYG; funded accounts; transitional dynamics; distributional welfare. **JEL:** E21; H55; J26; J46.

1 Introduction

Workers in economies with large informal sectors choose both whether to enter formal employment and how much value they obtain from mandatory pension contributions. Under Colombia’s Law 100 of 1993, formal workers could choose between a PAYG regime and an individual funded account ([Congreso de Colombia, 1993](#)). Law 2381 of 2024 replaces that competition with complementarity: every formal worker contributes to PAYG on earnings up to 2.3 statutory minimum wages, while only the excess enters the funded component ([Congreso de Colombia, 2024](#)). The reform therefore changes the choice set itself, not merely contribution rates.

This paper asks three questions. How does that institutional change affect informality, capital, wages, taxes, and pension participation? Which cohorts gain or lose during the transition? How are gains and losses distributed across worker types within each generation? These are counterfactual questions. The Constitutional Court suspended the integral entry into force of Law 2381 while it reviews the corrected legislative procedure ([Corte Constitucional de Colombia, 2025](#)); the experiment below is consequently not an observed before-and-after evaluation.

I answer the questions in a two-period overlapping-generations economy. One period represents roughly 40 years. Heterogeneous workers choose informality or formality, voluntary saving, and, when Law 100 applies, one of two pension regimes. A funded balance is portable and earns the market return. PAYG eligibility increases with worker type; an ineligible contributor receives a refund, while an eligible contributor obtains a defined benefit with a floor and a ceiling. These differences produce the benchmark ordering

$$\text{informal} \longrightarrow \text{capitalization} \longrightarrow \text{PAYG}$$

without quotas or random-utility shocks. Under Law 2381, the second choice disappears.

A formal worker receives a single package containing PAYG and, above the statutory threshold, funded saving. The endogenous allocation is therefore among informality, PAYG-only formality, and mixed PAYG–funded formality.

The legal crosswalk matters quantitatively. The total contribution is 16 percent. Thirteen points on earnings up to 2.3 minimum wages finance the common PAYG fund, and 13.2 points on the excess are credited to the individual account. The PAYG replacement rate is $65.5 - 0.5s$ percent, where s is the covered income in minimum wages, and its contributory benefit cannot fall below one minimum wage. The solidarity benefit is normalized to the 2023 extreme-poverty line, $218,846/1,160,000 = 0.1887$ minimum wages, and its targeted mass is proxied by the 12.6 percent incidence of extreme poverty ([Departamento Administrativo Nacional de Estadística, 2025](#); [Presidencia de la Republica de Colombia, 2022](#)).

The Law 100 benchmark has 50.9 percent informality, 46.0 percent capitalization, and 3.1 percent PAYG. The central pillar steady-state comparison has approximately 77.5 percent informality, 6.2 percent PAYG-only formality, and 16.3 percent mixed formality. Its consumption tax is about 53.4 percent. Those magnitudes are not forecasts: the 40-year conversion of a PAYG flow is not identified. A stress value that transfers the benchmark multiplier of 2.9 to the pillar formula raises the stationary tax to about 74.6 percent and is reported only as a stationary stress case.

The main result concerns dynamics rather than the terminal comparison. The perfect-foresight path now converges to the pillar steady state. The first cohort begins with a 17.6 percent consumption tax; the tax rises to 64.7 percent for cohort 2 as preserved Law 100 obligations come due, then its deviations decay. Public debt ranges from a government asset position of 0.42 percent of output to debt of 0.08 percent and is effectively zero by the end of the horizon. Informality and capital converge with the same damped pattern. The mechanism remains intergenerational—formal contributions today create PAYG obligations tomorrow—but debt absorbs part of the contemporaneous gap and the tax rule reacts before that gap is passed wholesale to the next generation. The first reform cohort loses 5.5 percent in consumption-equivalent terms; cohort 2 loses 2.0 percent and the loss converges to 3.3 percent.

The paper contributes an endogenous institutional mapping, a transition experiment that does not impose convergence without checking the terminal equations, an explicit debt-and-tax closure, and a distributional accounting of wages and welfare within and across cohorts. The rest of the paper reviews the literature, presents the model, maps both laws into parameters, reports stationary and transition results, and details the

algorithm and its validation.

2 Related literature

The model follows the overlapping-generations approach to social security in [Diamond \(1977\)](#). Mandatory retirement programs can address undersaving ([Feldstein, 1985](#)), while present-biased preferences provide a behavioral rationale for mandatory contributions ([Laibson, 1998](#); [İmrohoroglu et al., 2003](#)). [Pardo \(2023\)](#) introduces mandatory retirement saving into an economy with endogenous informality. The present paper adds pension-institution choice under Law 100 and then removes that choice under the mandatory pillar package. This permits a transition analysis in which the policy changes both returns and the feasible alternatives.

Pension design cannot be separated from labor informality. Payroll-financed programs alter the value of formal employment ([Levy, 2008](#)), and informality reflects choices by workers and firms rather than a residual labor-market state ([Ulyssea, 2018](#)). Search frictions and human-capital accumulation create heterogeneous formal careers ([Bobba et al., 2021](#)). The shadow economy also changes the information and tax bases available for redistribution ([Doligalski and Rojas, 2023](#)).

Quantitative pension models with informality include [Joubert \(2015\)](#) and [McKiernan \(2020\)](#). Related work studies retirement and occupational behavior when formal contributions can be avoided ([Ferreira and Parente, 2020](#); [Moreno, 2022](#)). The mechanism studied here is the interaction between pension eligibility, contribution portability, and the formal–informal margin. The stationary benchmark identifies conditions for interior regime coexistence; the reform experiment then studies how mandatory complementarity propagates through fiscal closure, wages, and cohort welfare. The debt experiment shows that the oscillation under strict contemporaneous balance is not a structural prediction of pension design: it depends on whether the fiscal authority can transfer a bounded part of the adjustment across cohorts.

For Colombia, [Becerra \(2025\)](#) develops the CEDE Pension Model, a dynamic microsimulation that combines household surveys, administrative records, legal rules, and population projections. That model supplies the institutional rates and empirical anchors used below. Its annual partial-equilibrium accounting and the present model’s 40-year general-equilibrium period are not interchangeable. The calibration therefore uses a documented crosswalk rather than copying the entire parameter vector.

3 Model

3.1 Demography, heterogeneity, and preferences

One model period represents approximately 40 years. Population grows at rate n , labor-augmenting productivity at rate g , and a worker survives to old age with probability $1 - m$. Type $i \in [0, 1]$ has density

$$f(i) = \frac{2.4450}{1.006341403957648} \exp \left[- \left(\frac{i - 0.4655}{0.2329} \right)^2 \right], \quad \int_0^1 f(i) \, di = 1. \quad (1)$$

Sector-specific efficiency is

$$z_f(i) = \text{base}_f e^{\rho_f i}, \quad z_s(i) = \text{base}_s e^{\rho_s i}. \quad (2)$$

Instantaneous utility is CRRA, $u(c) = c^{1-\sigma}/(1-\sigma)$. Decision utility and experienced utility are

$$V_j(i) = u(c_{1j}(i)) + \beta(1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (3)$$

$$W_j(i) = u(c_{1j}(i)) + (1 - m)\delta u(\tilde{c}_{2j}(i)), \quad (4)$$

where $j \in \{I, C, P\}$ denotes informal employment, capitalization, and PAYG. The parameter $\beta < 1$ affects saving and regime choice but is not used as a normative welfare weight.

3.2 Technology and household budgets

Formal and informal output are

$$Y_f = K^\alpha N_f^{1-\alpha}, \quad Y_s = A_s N_s. \quad (5)$$

Competitive pre-tax prices satisfy

$$\tilde{r} + \delta_k = \alpha(N_f/K)^{1-\alpha}, \quad \tilde{w}_f = (1 - \alpha)(K/N_f)^\alpha, \quad \tilde{w}_s = A_s. \quad (6)$$

Taxes map these into the net return r and the effective wages w_f and w_s .

For choice j , voluntary saving $a_j(i) \geq 0$ satisfies

$$(1 + \tau_c)c_{1j}(i) + (1 + g)a_j(i) = y_{1j}(i), \quad (7)$$

$$(1 - m)(1 + \tau_c)c_{2j}(i) = (1 + r)a_j(i) + p_j(i). \quad (8)$$

Informal income is $y_{1I}(i) = w_s z_s(i)$. Both formal alternatives have

$$y_{1C}(i) = y_{1P}(i) = (1 - \tau_\ell - \tau_p) w_f z_f(i). \quad (9)$$

Both formal alternatives pay the statutory rate τ_p . The funded account credits $\tau_C \leq \tau_p$, while the PAYG notional balance credits $\tau_N \leq \tau_p$; uncredited amounts enter consolidated public revenue. The regimes therefore share the current wedge but differ in credited balances and retirement benefits. Baseline transfers are zero.

3.3 Capitalization

A funded contribution is fully vested and portable:

$$A_C(i) = \tau_C w_f z_f(i), \quad (10)$$

$$p_C(i) = \frac{(1 + r)A_C(i)}{1 + g}. \quad (11)$$

The contribution adds to the capital stock. The worker receives the accumulated value regardless of whether the subsequent formal career would have satisfied PAYG vesting requirements.

3.4 PAYG eligibility, refund, and pension formula

The probability that type i completes the PAYG eligibility requirement is

$$q(i) = \frac{\exp(\gamma_0 + \gamma_1 i)}{1 + \exp(\gamma_0 + \gamma_1 i)}, \quad \gamma_1 > 0. \quad (12)$$

The pension conditional on eligibility is

$$\mathcal{B}(y_i) = \min \{B_{\max}, \chi \max [B_{\min}, by_i]\}, \quad y_i = w_f z_f(i), \quad (13)$$

where b is the replacement parameter and χ converts the replacement flow into units of the long model period. It also captures the implicit subsidy embedded in the PAYG formula.

If the worker fails to qualify, a fraction φ of the notional balance is returned. Let r_N denote the annual real return credited to that balance. Expected PAYG retirement income is therefore

$$p_P(i) = q(i) \frac{(1-m)\mathcal{B}(y_i)}{1+g} + [1-q(i)]\varphi \frac{(1+r_N)^{40}\tau_N y_i}{1+g}. \quad (14)$$

In the Colombian calibration, $\varphi = 1$ and $r_N = 0$: a non-eligible worker recovers the notional contribution adjusted for inflation but receives no real return. Expected PAYG benefits, including refunds, are public expenditure.

3.5 Joint choice and endogenous thresholds

Every worker solves the saving problem under all three alternatives and chooses

$$j^*(i) = \arg \max_{j \in \{I, C, P\}} V_j(i). \quad (15)$$

An interior three-regime equilibrium has thresholds $0 < i_1 < i_2 < 1$ satisfying

$$V_I(i_1) = V_C(i_1) > V_P(i_1), \quad (16)$$

$$V_C(i_2) = V_P(i_2) > V_I(i_2). \quad (17)$$

The resulting allocation is

$$j^*(i) = \begin{cases} I, & i < i_1, \\ C, & i_1 \leq i < i_2, \\ P, & i \geq i_2. \end{cases} \quad (18)$$

Equation (18) is an equilibrium outcome. The algorithm evaluates all three utilities point by point and does not impose either threshold.

3.6 The pillar reform

Law 2381 changes both the alternatives and the contribution bases. Let $w_{\min}$ denote one minimum wage in model units and $\bar{y} = 2.3w_{\min}$. For a formal worker with earnings

y_i , the PAYG and funded bases are

$$y_i^P = \min\{y_i, \bar{y}\}, \quad (19)$$

$$y_i^C = \max\{y_i - \bar{y}, 0\}. \quad (20)$$

The worker pays the 16 percent total contribution on both bases. Thirteen percent of y_i^P enters the common PAYG fund; 13.2 percent of y_i^C is credited to the individual account. The remaining shares finance administration, insurance, and the public savings fund, which are consolidated in the government budget.

For covered earnings measured in minimum wages, $s_i = y_i^P/w_{\min}$, the statutory PAYG rate is

$$b^{2381}(s_i) = 0.655 - 0.005s_i. \quad (21)$$

The modeled defined benefit is

$$\mathcal{B}^{2381}(y_i^P) = \chi_{2381} \max\{w_{\min}, b^{2381}(s_i)y_i^P\}, \quad (22)$$

subject to the statutory maximum. The central experiment sets $\chi_{2381} = 1$ because the long-period conversion is not legally identified; $\chi_{2381} = 2.9$, the value selected for the Law 100 interior branch, is reported only as a stress test.

The solidarity transfer is targeted to the lowest types among informal workers. With cutoff i_S chosen so that $F(i_S) = 0.126$,

$$S(i) = \mathbf{1}\{j^*(i) = I, i \leq i_S\}, \quad B^S = 0.1887w_{\min}. \quad (23)$$

This is a parsimonious proxy for poverty targeting. It does not separately represent the semicontributory pillar, contribution histories, disability, sex-specific ages, or household means tests.

Formal workers now choose a package rather than an exclusive pension regime:

$$j_{2381}^*(i) = \arg \max_{j \in \{I, F\}} V_j(i). \quad (24)$$

A formal worker with $y_i \leq \bar{y}$ is classified as PAYG-only, while one with $y_i > \bar{y}$ is classified as mixed PAYG–funded. Thus positive masses in the three reported categories arise from the endogenous formality threshold and the statutory earnings threshold; they are not three pension-plan choices under the new law.

3.7 Aggregation and fiscal closure

Let $d_j(i) = \mathbf{1}\{j^*(i) = j\}$. Regime shares and effective labor are

$$\pi_j = \int_0^1 d_j(i) f(i) di, \quad (25)$$

$$N_f = \int_0^1 [d_C(i) + d_P(i)] z_f(i) f(i) di, \quad (26)$$

$$N_s = \int_0^1 d_I(i) z_s(i) f(i) di. \quad (27)$$

Gross household assets carried to the next period are

$$S' = \frac{1}{1+n} \int_0^1 a_{j^*(i)}(i) f(i) di + \frac{1}{(1+n)(1+g)} \int_0^1 d_C(i) \tau_C w_f z_f(i) f(i) di. \quad (28)$$

Government bonds and productive capital are imperfectly integrated. If B' is public debt per efficiency unit of the incoming cohort and ω_B is the share absorbed domestically, asset-market clearing is

$$K' = S' - \omega_B B'. \quad (29)$$

The remaining share is placed externally. This reduced-form opening is necessary because one model period spans a generation; it is not an estimate of Colombia's investor base.

Public pension revenue is

$$R^{pen} = \int_0^1 [d_P(i) \tau_p + d_C(i) (\tau_p - \tau_C)] w_f z_f(i) f(i) di. \quad (30)$$

Write R_t^{nc} for labor, payroll, capital, and pension revenue. Write E_t for government spending, transfers, solidarity benefits, and PAYG obligations. The consumption-tax base is $\mathcal{C}_t$. The primary balance is

$$PB_t = \tau_{ct} \mathcal{C}_t + R_t^{nc} - E_t. \quad (31)$$

Public debt obeys

$$(1+n)(1+g)B_{t+1} = (1+r_t)B_t - PB_t. \quad (32)$$

For inherited debt B_t , define the tax that would place next period's debt at target $\bar{B}$,

$$\tau_{ct}^{stab} = \frac{E_t - R_t^{nc} + (1+r_t)B_t - (1+n)(1+g)\bar{B}}{\mathcal{C}_t}. \quad (33)$$

The actual tax is bounded by a logit map. Let $z(x) = \log[x/(\tau^{max} - x)]$, let Λ be the logistic function, and truncate τ_{ct}^{stab} to the open interval $(\varepsilon, \tau^{max} - \varepsilon)$. The fiscal rule is

$$\tau_{ct} = \tau^{max} \Lambda \left[\omega_{\tau} z(\bar{\tau}_c) + (1 - \omega_{\tau}) z(\tilde{\tau}_{ct}^{stab}) \right]. \quad (34)$$

Thus ω_{τ} is a long-run tax-anchor weight, not a delayed reaction coefficient. When $B_t = \bar{B}$ and the allocation is stationary, the rule returns $\bar{\tau}_c$; away from that point, debt absorbs the part of the fiscal gap not closed by the contemporaneous tax.

Under pillars, the funded term in gross asset accumulation becomes

$$\frac{\int \mathbf{1}\{j^* = F\} 0.132 y_i^C f(i) di}{(1+n)(1+g)}.$$

Public pension revenue becomes

$$R_{2381}^{pen} = \int_0^1 \mathbf{1}\{j^*(i) = F\} \left[0.16 y_i^P + (0.16 - 0.132) y_i^C \right] f(i) di. \quad (35)$$

The PAYG benefit, refunds, the solidarity transfer, and preserved Law 100 rights of the old cohort enter E_t . Equations (32)–(34) make debt and the consumption tax joint intergenerational state variables.

4 Calibration

4.1 Parameter crosswalk

The Colombian inputs come from the CEDE Pension Model documented by [Becerra \(2025\)](#). That source is an annual dynamic microsimulation; the present model is a stationary 40-year OLG equilibrium. I therefore classify each input as a direct legal parameter, a comparable calibration moment, an external validation, or a retained structural parameter. This prevents annual accounting assumptions from being reinterpreted as balanced-growth parameters.

For the Law 100 benchmark, the direct legal inputs are the 16 percent total contribution rate, the 11.5 percent credited to an individual account, and the 13 percent credited to the PAYG notional balance. A funded worker pays the full 16 percent: the credited balance becomes capital and the remainder enters consolidated public revenue. A PAYG worker also pays 16 percent, while a failed eligibility draw returns the 13 percent notional balance with no real interest. The replacement parameter is 0.65 near one

minimum wage.

The reform crosswalk is separate. Under Law 2381, 13 points below the 2.3-minimum-wage threshold enter the common PAYG fund, while 13.2 points above it enter the funded account. The remaining 2.8 points on the excess consolidate administration, insurance, and public saving. The legal replacement formula and one-minimum-wage floor enter directly. The solidarity amount uses the 2023 extreme-poverty line, while the share of low types eligible for targeting uses the national 12.6 percent extreme-poverty incidence. These are a benefit ratio and a targeting proxy, respectively; neither should be interpreted as an estimate of take-up among older adults.

Table 1 reports the complete benchmark. Annual population growth, productivity growth, and discounting remain compounded over 40 years: $n = 1.013^{40} - 1$, $g = 1.004^{40} - 1$, and $\beta = 0.97^{40}$. The source assumes 1.6 percent annual real wage growth, but that rate only indexes earnings and benefits in its microsimulation. Substituting it for balanced-growth productivity g changes every intergenerational scaling equation and does not yield a stable benchmark. Likewise, the source’s 4 percent real return is retained as an external check rather than imposed on the endogenous return of a closed economy.

Table 1: Baseline parameters and provenance

Parameter	Value	Use	Interpretation
m	0.4000	retained	mortality before old age
n	0.6764	retained	population growth per model period
σ	2.5000	retained	CRRA curvature
β	0.2957	retained	behavioral discount factor
g	0.1731	retained	balanced-growth productivity
α	0.4000	retained	capital share
A_s	0.08215	calibrated	matches the long-run formality anchor
ρ_s, ρ_f	2.90, 4.64	retained	type gradients
τ_ℓ, τ_n	0.125, 0.125	retained	labor and payroll rates
τ_p	0.1600	direct	total pension contribution
τ_C	0.1150	direct	credited funded-account rate
τ_N	0.1300	direct	credited PAYG notional rate
r_N	0.0000	direct	real return on a PAYG refund
τ_k	0.3200	retained	capital-income tax
b	0.6500	direct	replacement rate near one minimum wage
G	0.3300	retained	government purchases
γ_0	-8.7900	calibrated	matches the eligibility proxy
γ_1	15.0000	retained	shape normalization for eligibility
φ	1.0000	direct	refund fraction if not eligible
χ	2.9000	selected	40-year benefit conversion; interior branch
$B_{\min}$	0.0500	retained	floor in internal OLG units
$B_{\max}$	100.0000	retained	nonbinding ceiling

Table 2: Law 2381 policy mapping

Parameter		Value	Status
Total contribution rate		0.1600	direct legal input
PAYG credit below threshold		0.1300	direct legal input
Funded credit above threshold		0.1320	direct legal input
PAYG earnings threshold	2.3000	SMLMV	direct legal input
Replacement intercept		0.6550	direct legal input
Replacement slope		0.0050	direct legal input
Minimum contributory benefit	1.0000	SMLMV	direct legal input
Solidarity benefit	0.1887	SMLMV	calibration ratio
Solidarity target mass		0.1260	calibration proxy
χ_{2381} central (stress)	1.0000 (2.9000)		not identified

Notes: SMLMV denotes the statutory monthly minimum wage. The solidarity ratio is 218,846/1,160,000. The 2.9 multiplier is not a legal input.

4.2 Moments and external checks

The CEDE projection raises formal employment from 42 percent in 2020 to 48.5 percent in 2100. I use 48.5 percent as the stationary formality anchor. It also projects contributory coverage near 25 percent in 2080–2100. Because the OLG has no contribution-history state, I approximate conditional eligibility among formal workers by $0.25/0.485 = 0.51546$. This ratio is a calibration proxy, not an accounting identity: contributory coverage in the source includes DB, DC, and DB+DC pensions and is measured among older adults.¹

The two comparable moments discipline A_s and γ_0 . The 10 percent DB enrollment share among young entrants and the 4 percent real return in the source are not imposed. They are external checks because the former is a cohort-specific plan assignment and the latter is an actuarial assumption rather than a general-equilibrium price. Table 3 reports both the matched moments and these transparent discrepancies.

The source does not identify the logistic slope, the conversion from an annual pension flow to a 40-year benefit, or the minimum wage in the model’s efficiency-wage units. Accordingly, γ_1 , χ , and the wage normalization are not empirical estimates. The value $\chi = 2.9$ selects the interior Law 100 branch; it is not transferred to the central reform

¹The coverage rows in Table 2 of [Becerra \(2025\)](#) are labeled in the reverse order. Section 4.1, Figure 4, and Table 1 consistently report contributory coverage of 23.7–23.8 percent on average and non-contributory coverage of 53.1–53.3 percent. The calibration uses the separately reported long-run contributory value of approximately 25 percent.

Table 3: Calibration moments and external validation				
Use	Moment	Target	Model	Gap
Calibrated	Formal share	0.48500	0.49081	0.00581
Calibrated	Eligibility among formals	0.51546	0.51373	−0.00173
Validation	PAYG share among formals	0.10000	0.06353	−0.03647
Validation	Annual real return	0.04000	0.05741	0.01741

Notes: The eligibility target is the proxy 0.25/0.485. The PAYG validation uses the approximately 10 percent DB enrollment share among entrants aged 20–30 reported for 2017.

experiment. The reported stationary reform solution is refined to 201 types, while the transition uses 101 diagnostic types plus exact root insertion at every choice boundary.

The fiscal-transition parameters are not identified by the CEDE microsimulation. Table 4 therefore labels them as provisional. The zero debt target is a normalization, the 5 percent tax-anchor weight leaves 95 percent of the logit response to the contemporaneous stabilizing tax, and $\omega_B = 0.25$ means that one quarter of public debt displaces domestic productive capital. The 90 percent ceiling defines the numerical policy domain; it is not a proposed tax.

Table 4: Provisional fiscal-transition parameters		
Parameter	Interpretation	Value
$\bar{B}$	Public-debt target	0
ω_τ	Long-run tax-anchor weight	0.05
ω_B	Domestic share of public debt	0.25
τ^{max}	Upper bound of the fiscal-rule domain	0.90

Notes: Values are reduced-form transition assumptions, not empirical estimates. One model period is approximately 40 years, so they should not be interpreted as annual fiscal-rule coefficients.

5 Results

5.1 Benchmark equilibrium

Table 5 reports the refined stationary equilibrium. All three alternatives have positive mass. Informality represents 50.9 percent of workers, capitalization 46.0 percent, and PAYG 3.1 percent. PAYG therefore accounts for 6.4 percent of formal workers. The

formal-entry threshold is $i_1 = 0.4696$ and the pension-regime threshold is $i_2 = 0.7713$.

Table 5: Endogenous-choice steady state

Outcome	Value	Outcome	Value
Informal share	0.50919	Capital stock	0.02968
Capitalization share	0.45962	Formal labor	8.89024
PAYG share	0.03118	Informal labor	1.40926
First threshold i_1	0.46965	Consumption tax	0.19612
Second threshold i_2	0.77128	Annualized return	0.05741
Young consumption	0.39523	Formal wage	0.05450
Old-age consumption	0.27038	Voluntary capital	0.00620
Funded capital	0.02348	PAYG contributions	0.01328
Decision welfare	-5.43803	PAYG benefits	0.04661
Experienced welfare	-5.93741	Maximum residual	1.8×10^{-7}

The first threshold reflects the formal tax wedge. Types below 0.4696 obtain too little formal-sector income to compensate for labor and pension contributions. Above that threshold, formal productivity grows faster than informal productivity and capitalization becomes optimal. PAYG does not attract the middle group because its probability of completing the eligibility requirement is still too low.

The second threshold reflects the interaction between eligibility and the implicit PAYG subsidy. At $i = 0.7713$, the expected PAYG benefit overtakes the portable funded balance. Higher types remain in PAYG because eligibility continues to increase and the benefit ceiling does not bind. The two thresholds therefore arise from different comparisons: $V_C - V_I$ at the first threshold and $V_P - V_C$ at the second.

PAYG benefits are 3.51 times contemporaneous PAYG contributions. Part of this difference reflects demographic scaling and part reflects χ , which is required to compare an annual replacement flow with a 40-year funded balance. The consolidated budget finances the gap. This is why regime choice cannot be analyzed independently of fiscal closure. The model's 6.4 percent PAYG share among formals is below the 10 percent enrollment check from young entrants. Closing that gap mechanically by raising χ moves the equilibrium toward a high-tax PAYG branch, rather than providing an innocuous fit.

5.2 Eligibility sensitivity

Table 6 changes only the eligibility intercept. Increasing γ_0 makes every type more likely to qualify while preserving the slope of the profile. Positive masses remain in all three alternatives throughout the reported range.

Table 6: Sensitivity to PAYG eligibility

	$\gamma_0 = -8.90$	Benchmark	$\gamma_0 = -8.70$
Informal share	0.48269	0.50929	0.54806
Capitalization share	0.49050	0.45951	0.41403
PAYG share	0.02681	0.03120	0.03792
Consumption tax	0.17123	0.19622	0.23667
First threshold i_1	0.45874	0.46969	0.48570
Second threshold i_2	0.78201	0.77125	0.75696

Notes: Sensitivity equilibria use 201 types and the refined benchmark as their initial state. All three converge and have maximum fixed-point residuals below 4.1×10^{-6} .

Easier eligibility lowers the second threshold and raises PAYG participation, as expected. It also increases expected benefits and the consumption tax. The higher tax raises the first threshold, so informality increases by 6.5 percentage points between the two endpoints. The capitalization share falls by 7.6 points: some workers enter PAYG while others leave formality. This is a general-equilibrium implication of the calibrated fiscal rule, not a claim that easier eligibility must raise informality in every pension system.

5.3 Stationary pillar comparison

Table 7 compares the two stationary allocations. The pillar system does not produce an exclusive funded category. About 6.2 percent of workers are formal and below the statutory threshold, while 16.3 percent are formal with both PAYG and funded components. Informality rises to approximately 77.5 percent and the consumption tax to 53.4 percent. The result reflects the consolidated fiscal rule, the PAYG floor, and the solidarity transfer; it is not a forecast of take-up or tax legislation.

The stress experiment transfers $\chi = 2.9$ from the Law 100 sorting calibration to the pillar benefit. It produces roughly 79 percent informality and a consumption tax near 75 percent in the stationary comparison. I do not use that underidentified multiplier in the

Table 7: Stationary comparison: Law 100 and central pillars

Outcome	Law 100	Pillars
Informal share	0.5093	0.7745
Funded-only share	0.4595	0.0000
PAYG-only share	0.0312	0.0622
Mixed PAYG–funded share	0.0000	0.1633
Consumption tax	0.1962	0.5334
Capital stock	0.0297	0.0250
Formal wage	0.0545	0.0604

Notes: Values shown use the reproducible reform script. The Law 100 table above uses the more refined benchmark script, which explains immaterial rounding differences.

central transition. The stress result is evidence against treating $\chi = 2.9$ as a portable legal parameter, not proof that the statutory reform is infeasible.

5.4 Transitional dynamics with debt and a fiscal rule

The reform begins with cohort 1. The old at that date retain their Law 100 contributory rights but can receive the new solidarity transfer. Cohort 1 and every subsequent entrant face the pillar rules. Figure 1 reports the complete 20-cohort computational horizon.

The first reform cohort has 55.7 percent informality and a 17.6 percent consumption tax. Its primary surplus creates a government asset position inherited by cohort 2. When preserved Law 100 benefits come due, cohort 2 faces a 64.7 percent tax, 81.5 percent informality, and a capital stock of 0.0238. The rule then reverses part of the adjustment: cohort 3 has a 45.6 percent tax and 74.5 percent informality. Subsequent deviations shrink rather than repeating at constant amplitude.

Table 8: Selected points on the validated transition

Outcome	Law 100	Cohort 1	Cohort 2	Cohort 5	Steady state
Informal share	0.5093	0.5568	0.8152	0.7657	0.7745
PAYG-only share	0.0312	0.3052	0.0121	0.0732	0.0622
Mixed share	0.0000	0.1380	0.1727	0.1611	0.1633
Consumption tax	0.1962	0.1763	0.6470	0.5094	0.5334
Capital stock	0.0297	0.0297	0.0238	0.0252	0.0250
Debt/output (percent)	0.0000	0.0000	−0.4180	0.0336	0.0000
Primary balance	0.0000	0.0069	−0.0287	0.0028	0.0000

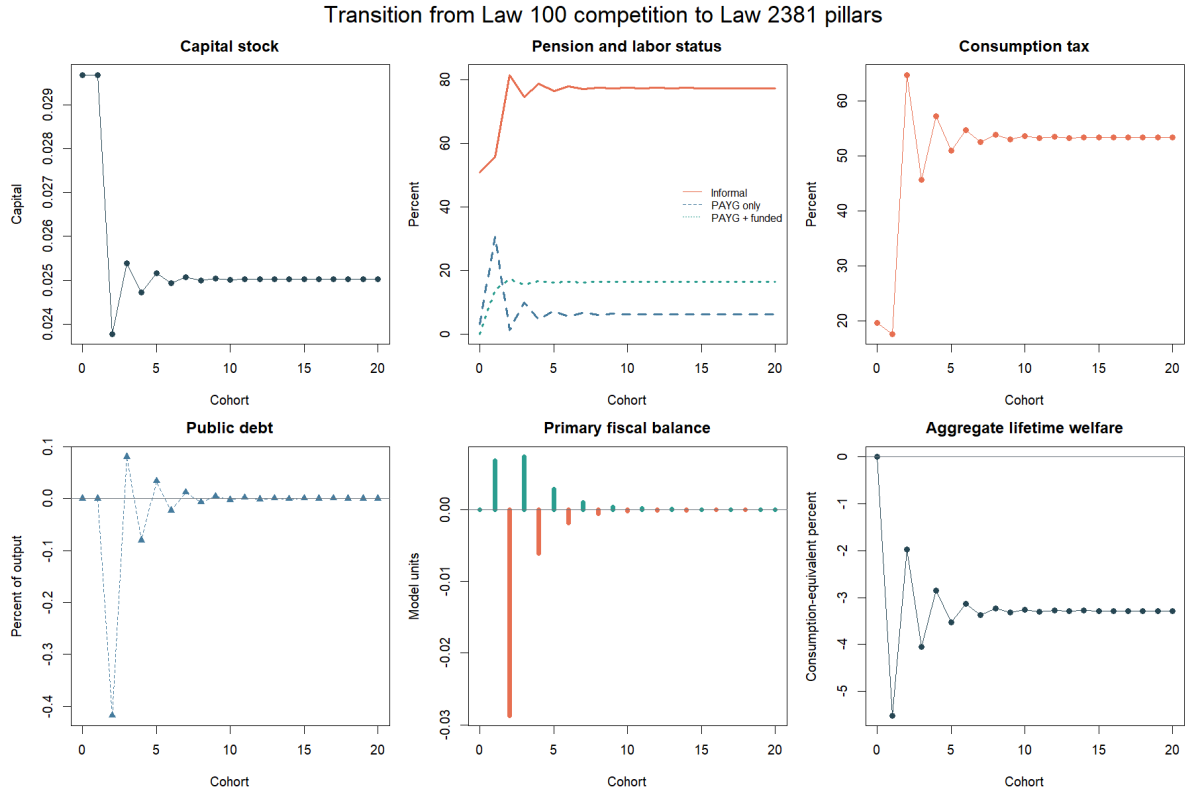


Figure 1: Endogenous transition from Law 100 to the pillar system
Notes: One cohort is one model period of approximately 40 years. Cohort 0 is the Law 100 benchmark and cohort 20 is the stationary pillar endpoint. Negative public debt denotes government assets. The primary balance excludes interest payments.

The same fiscal lag that generated the earlier accordion remains economically active: higher formality raises contributions today and PAYG obligations tomorrow. What changes is fiscal timing. Equation (34) responds to the current debt-stabilizing tax, while equation (32) carries only the residual across cohorts. By cohort 13, the tax is 53.29 percent, informality 77.43 percent, and debt less than 0.001 percent of output. The joint terminal gap is 8.5×10^{-5} , the debt-identity residual is zero to machine precision, and the maximum fiscal-rule residual is 1.93×10^{-4} . A two-cycle is no longer an accepted validation alternative.

5.5 Wage and welfare distributions

Figures 2 and 3 show that averages conceal substantial redistribution across types. The earnings distribution shifts back and forth with the fiscal adjustment at first and then converges. The welfare profiles retain pronounced type-specific kinks at the informality and pension thresholds even when aggregate variables are close to their steady state. These are equilibrium labor earnings, not observed Colombian wage pesos.

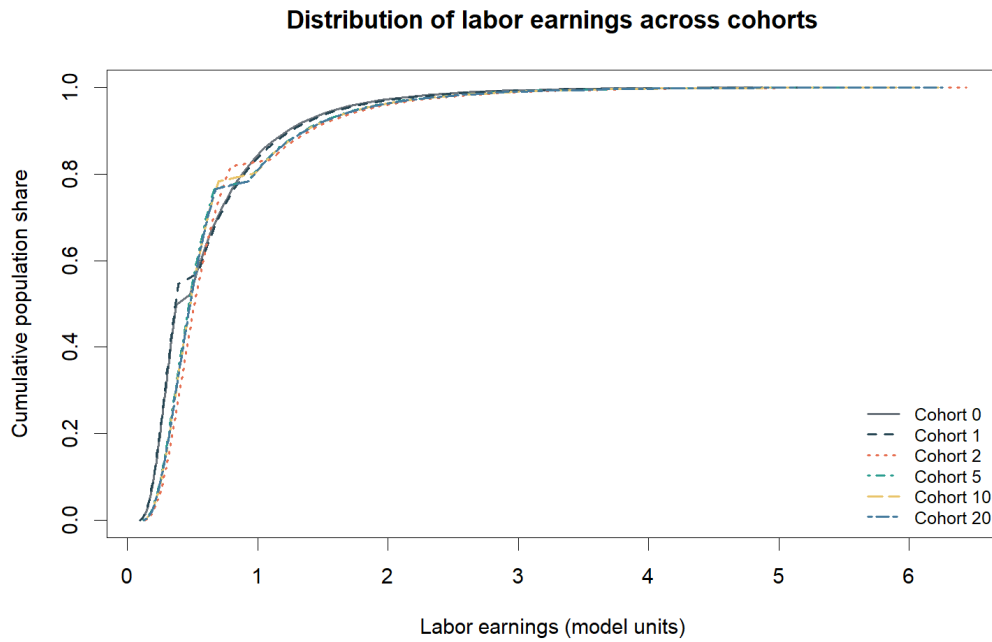


Figure 2: Distribution of labor earnings across selected cohorts

The old at the reform gain 11.8 percent in aggregate consumption-equivalent welfare because acquired contributory rights are preserved while the targeted solidarity benefit

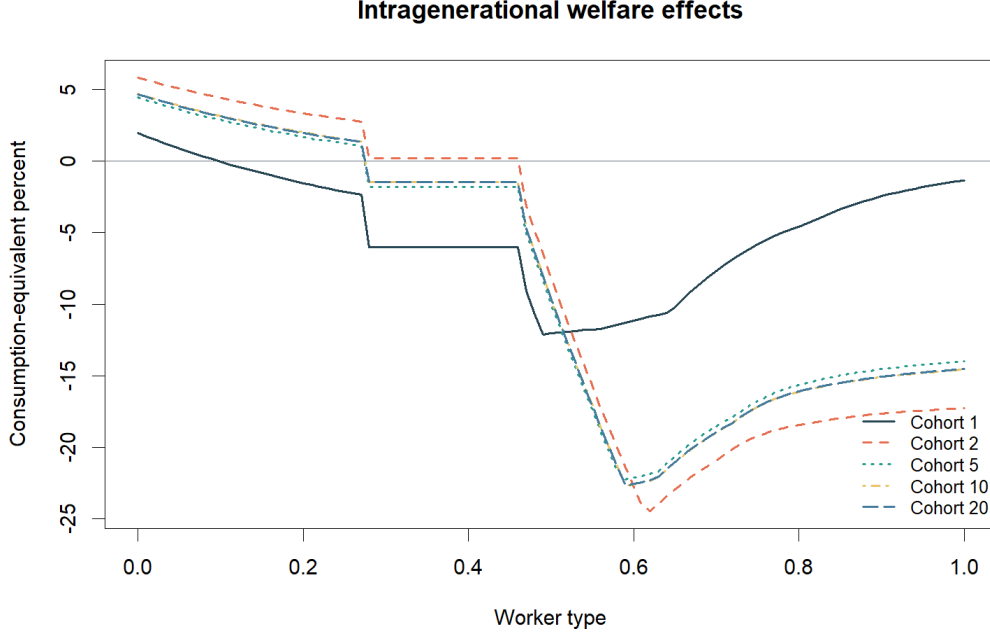


Figure 3: Intragenerational welfare effects by worker type

begins immediately. The gain is not broad based: only 15.1 percent gain, the median change is -1.0 percent, and the ninetieth percentile gains 36.9 percent. The first reform cohort loses 5.5 percent in aggregate and 99.0 percent of types lose. Cohort 2 loses 2.0 percent in aggregate; 49.8 percent gain, but the median loses 3.2 percent and the tenth percentile loses 22.5 percent. At the stationary pillar allocation, aggregate welfare is 3.3 percent below Law 100, 12.2 percent of types gain, the tenth percentile loses 21.3 percent, and the ninetieth percentile gains 1.4 percent. The transition therefore converges without becoming distributionally representative.

6 Conclusion

Moving from competing regimes to a pillar system changes more than pension assignment. Under Law 100, informality, capitalization, and PAYG coexist because portability and eligibility create comparative advantage across worker types. Under Law 2381, every formal worker enters PAYG up to 2.3 minimum wages and only higher earnings generate funded saving. The relevant endogenous categories become informal, formal PAYG-only, and formal mixed.

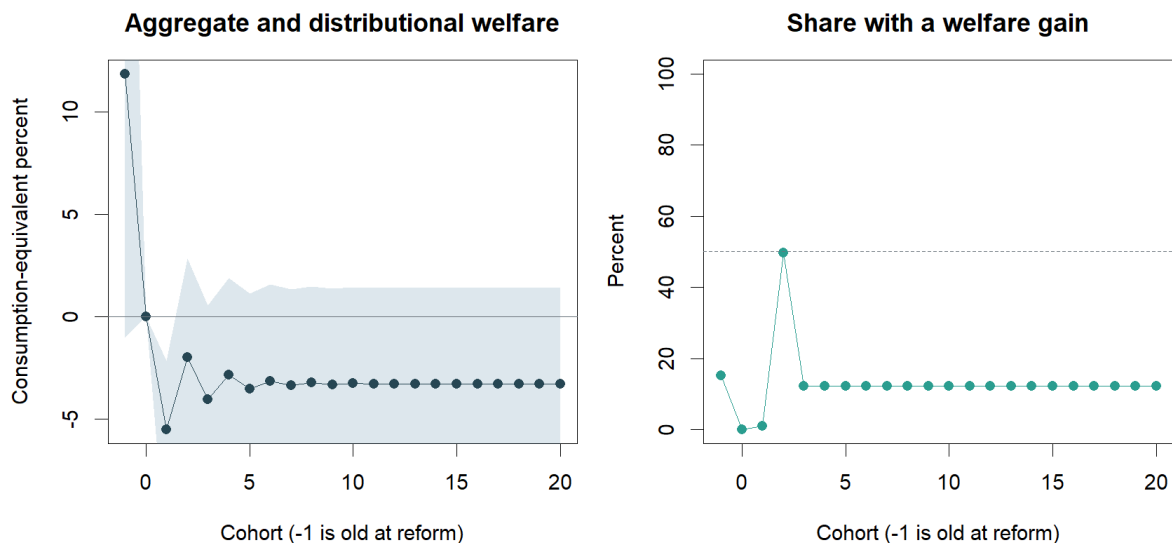


Figure 4: Intergenerational welfare and the share of winners

Notes: Welfare is computed with experienced utility, excluding the present-bias parameter from the normative weight. The shaded area spans the tenth to ninetieth percentile of individual consumption-equivalent changes.

The central stationary comparison raises informality and the consumption tax while preserving positive masses in all three categories. Under strict period-by-period balance, the delayed PAYG obligation generates a large two-generation oscillation. Once public debt and the contemporaneous fiscal rule are introduced, that oscillation is damped: capital, taxes, labor status, and debt converge to the pillar allocation. Fiscal closure therefore changes the dynamic result, not only the government's accounting.

Distributional accounting changes the policy interpretation. Preserving acquired rights and adding the solidarity benefit raises aggregate welfare for those already old when the reform begins, but the median old person loses. The first new cohort loses almost throughout the type distribution. Later cohorts converge to an aggregate loss near 3.3 percent while a small upper tail gains. A reform cannot therefore be summarized by one aggregate welfare number or by a stationary representative cohort.

Four cautions are essential. First, Law 2381 remains suspended, so the exercise is a counterfactual mapping of its statutory design. Second, one model period is approximately 40 years; the path is generational, not an annual forecast. Third, the debt target, tax-anchor weight, domestic debt share, and tax ceiling are provisional reduced-form assumptions, not estimated fiscal-rule coefficients. Fourth, the benefit multiplier, the mapping of the minimum wage into efficiency units, and contribution histories remain underidentified. The 2.9 multiplier used to obtain the Law 100 interior branch is

especially nonportable: transferring it to pillars produces an extreme fiscal stress case.

The next empirical step is to add age, sex, contribution histories, and the semicontributory rules from the CEDE microsimulation, and to value PAYG and funded annuities with a common actuarial conversion. A five-year multi-cohort version is also needed to calibrate debt maturity, fiscal adjustment, and domestic bond absorption at a meaningful frequency. The present result should be read as a mechanism and a discipline: one should neither accept an oscillatory path produced by an extreme fiscal closure nor impose a stationary endpoint without verifying the transition equations.

A Numerical algorithm

A.1 Household alternatives

For each candidate aggregate state (K, N_f, τ_c) , the algorithm computes prices and solves three household problems at every diagnostic type:

1. informal employment with no pension contribution;
2. formal employment with a fully funded contribution;
3. formal employment with PAYG eligibility, pension, and refund given by (12)–(14).

The saving solution imposes nonnegative voluntary saving and positive consumption. The algorithm records all three decision utilities and applies (15).

A.2 Locating endogenous thresholds

The diagnostic grid identifies every interval in which the selected alternative changes. Within each interval, a scalar root finder solves the relevant pairwise utility equation. The roots are added to the quadrature nodes. Choice is then evaluated at each interval midpoint and held constant only within that root-bounded interval. This procedure neither assumes the number of crossings nor imposes their order.

A.3 General-equilibrium fixed point

Given the exact choice intervals, trapezoidal quadrature aggregates labor, consumption, voluntary saving, funded contributions, PAYG flows, and welfare. The update map is

$$(K, N_f, N_s, \tau_c) \longmapsto (\widehat{K}, \widehat{N}_f, \widehat{N}_s, \widehat{\tau}_c), \quad (36)$$

where $\widehat{K}$ follows the capital-accumulation equation and $\widehat{\tau}_c$ solves the stationary consolidated budget. Damped updates continue until capital, both labor aggregates, the consumption tax, and the budget residual satisfy tolerance.

The implementation first solves 101 types and uses that state to initialize 501 types. Validation requires positive consumption, nonnegative saving, normalized regime shares, positive masses in all three choices, two interior roots, the sequence $I \rightarrow C \rightarrow P$, the income identity, and a maximum fixed-point residual below 2×10^{-6} . The sensitivity equilibria repeat the complete fixed point rather than holding prices or taxes fixed.

A.4 Transition path iteration

Let $T = 20$ be the computational horizon. The initial capital stock is predetermined by the Law 100 steady state and public debt starts at $B_0 = 0$. From cohort 1 onward, household problems use the pillar parameters. The old at the announcement date retain their contributory choice and accumulated saving from Law 100; the realized return and the new solidarity transfer determine their old-age consumption.

For a candidate sequence

$$\{K_t, N_{ft}, N_{st}, \tau_{ct}\}_{t=0}^T,$$

the algorithm performs four operations. First, it computes current prices at every date. Second, each young cohort solves its two alternatives using the candidate next-period return. Third, aggregation produces gross assets and implied labor sequences. Fourth, a forward fiscal sweep starts from B_0 . At each date it computes the debt-stabilizing tax, applies equation (34), obtains the primary balance, advances debt with equation (32), and maps gross assets into physical capital with equation (29). This triangular fiscal block prevents current debt from being an independent guessed path.

The real paths are updated with an old-iterate weight of 0.90 and the tax and tracked-debt paths with a weight of 0.95. These weights affect numerical convergence, not the fixed point. A saved validated path may be used as a warm start, but it is reevaluated under the current equations. The process stops only when the maximum relative change is below 10^{-4} and the fiscal-rule residual is below 2×10^{-4} . The pillar steady state supplies the boundary prices at cohort T ; it is not accepted as evidence that the preceding endogenous path converges.

Specifically, define the joint terminal gap

$$\Delta_T = \max \left\{ \frac{|\widehat{K}_T - \bar{K}|}{\bar{K}}, \frac{|\widehat{B}_T - \bar{B}|}{\bar{K}}, \frac{|\widehat{\tau}_{c,T-1} - \bar{\tau}_c|}{\bar{\tau}_c} \right\}. \quad (37)$$

A transition is validated only if the internal fixed point converges, $\Delta_T < 10^{-3}$, the maximum residual in the debt identity is below 2×10^{-4} , and the maximum residual in the fiscal rule is below the same tolerance. A cycle is not an admissible terminal outcome. In the central run, the respective terminal, debt-identity, and fiscal-rule residuals are 8.5×10^{-5} , zero to machine precision, and 1.93×10^{-4} . Figure 1 reports cohorts 0 through 20, including the fixed boundary point; convergence is established by the preceding tests rather than by that

A.5 Distributional accounting

The code retains 101 type-level observations for every cohort. Population weights are the quadrature weights of $f(i)$. It reports weighted wage quantiles and Gini coefficients, as well as the share of types with welfare gains. Normative welfare uses experienced utility W , not the present-biased decision utility V . For $\sigma \neq 1$, the consumption-equivalent change relative to the Law 100 allocation is

$$\lambda(i) = \left[\frac{W^{2381}(i)}{W^{100}(i)} \right]^{1/(1-\sigma)} - 1. \quad (38)$$

The old at reform receive an old-age-only equivalent; young cohorts receive a lifetime equivalent. Aggregate equivalents are computed from weighted aggregate utilities and are not the arithmetic mean of individual equivalents.

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